

Technical Data

Temperature

Temperature range:	+4 °C to +95 °C
Max. sterilisation temperature, dry steam:	140 °C (Max 30 min)

Physical Data

Materials

Product wetted parts:	1.4404 (316L)
Other steel parts:	1.4301 (304)
Seals:	EPDM
External finish:	Ra 1.5-2.5 µm
Internal finish:	Ra 0.8 µm
Connections:	Inlet: Liner/nut DIN 11851 Outlet: Male DIN 11851

Option:

Inductive sensor for feedback is available for standard and pneumatic lifting - see instruction manual for detail.

Dimensions (mm)

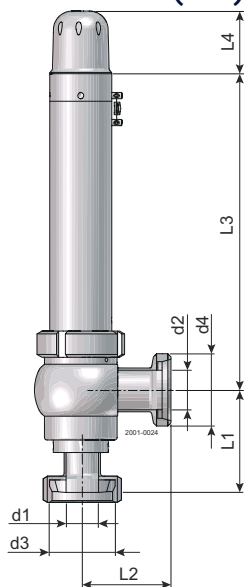


Figure 1. Standard DN25

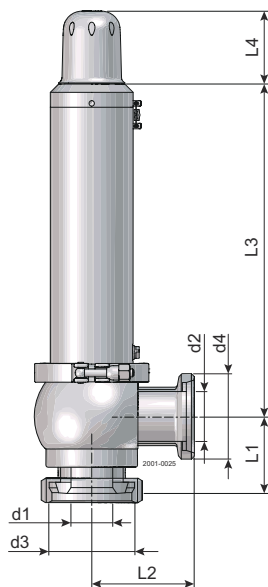


Figure 2. Standard DN40-DN100

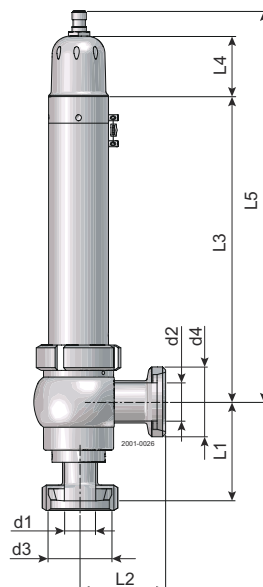


Figure 3. Standard DN25 with inductive sensor for feedback

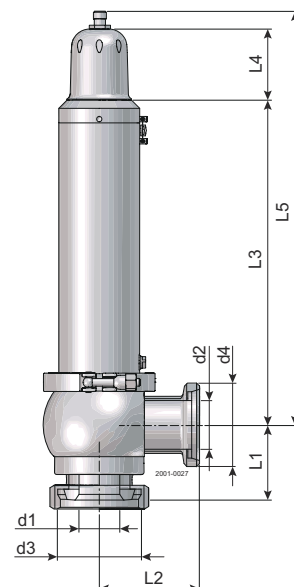


Figure 4. Standard DN40-DN100 with inductive sensor for feedback

Standard

Size	d1	d2	d3	d4	L1	L2	L3	L4	Kg
DN25	26	32	Rd52x1/6	Rd58x1/6	82	72	253	50	6.8
DN40	32	38	Rd65x1/6	Rd65x1/6	68	82	255	66	9.1
DN50	38	50	Rd78x1/6	Rd78x1/6	70	93	301	66	13.0
DN65	50	66	Rd95x1/6	Rd95x1/6	85	105	402	66	15.0
DN80	66	81	Rd110x1/4	Rd110x1/4	100	115	407.5	66	22.0
DN100	81	100	Rd130x1/4	Rd130x1/4	130	130	418	66	28.2

Standard with inductive sensor for feedback

Size	d1	d2	d3	d4	L1	L2	L3	L4	L5	Kg
DN25	26	32	Rd52x1/6	Rd58x1/6	82	72	253	50	324	6.8
DN40	32	38	Rd65x1/6	Rd65x1/6	68	82	255	66	338	9.1
DN50	38	50	Rd78x1/6	Rd78x1/6	70	93	301	66	384	13.0
DN65	50	66	Rd95x1/6	Rd95x1/6	85	105	402	66	484	15.0
DN80	66	81	Rd110x1/4	Rd110x1/4	100	115	407.5	66	489	22.0
DN100	81	100	Rd130x1/4	Rd130x1/4	130	130	418	66	501	28.2

Dimensions (mm)

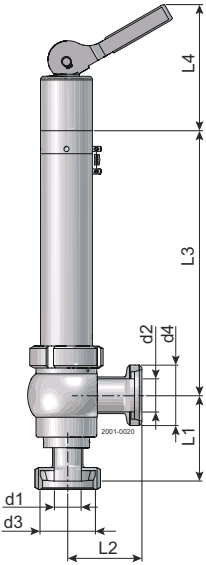


Figure 5. Manual lifting DN25

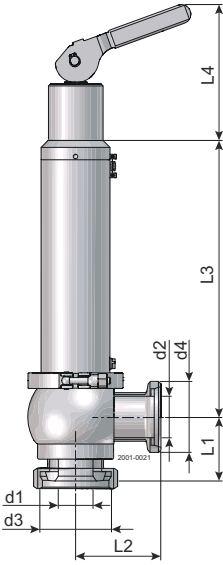


Figure 6. Manual lifting DN40-DN100

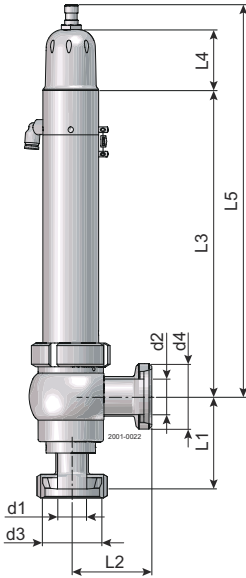


Figure 7. Pneumatic lifting DN25 with inductive sensor for feedback

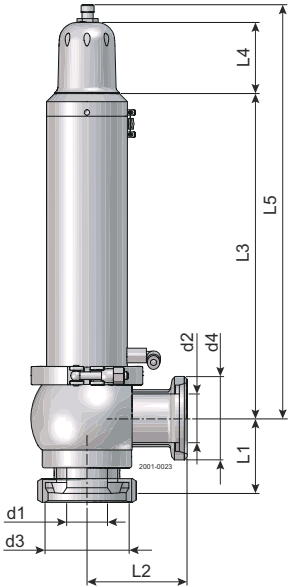


Figure 8. Pneumatic lifting DN40-DN100 with inductive sensor for feedback

Manual lifting

Size	d1	d2	d3	d4	L1	L2	L3	L4	Kg
DN25	26	32	Rd52x1/6	Rd58x1/6	82	72	253	141-182	7.5
DN40	32	38	Rd65x1/6	Rd65x1/6	68	82	255	152-232	10.3
DN50	38	50	Rd78x1/6	Rd78x1/6	70	93	301	154-234	15.5
DN65	50	66	Rd95x1/6	Rd95x1/6	85	105	402	153-233	16.2
DN80	66	81	Rd110x1/4	Rd110x1/4	100	115	407.5	152.5-232.5	23.2
DN100	81	100	Rd130x1/4	Rd130x1/4	130	130	418	152-232	29.6

Pneumatic lifting with inductive sensor for feedback

Size	d1	d2	d3	d4	L1	L2	L3	L4	L5	Kg
DN25	26	32	Rd52x1/6	Rd58x1/6	82	72	253	50	324	6.8
DN40	32	38	Rd65x1/6	Rd65x1/6	68	82	255	66	338	9.1
DN50	38	50	Rd78x1/6	Rd78x1/6	70	93	301	66	384	13.0
DN65	50	66	Rd95x1/6	Rd95x1/6	85	105	402	66	484	15.0
DN80	66	81	Rd110x1/4	Rd110x1/4	100	115	407.5	66	489	22.0
DN100	81	100	Rd130x1/4	Rd130x1/4	130	130	418	66	501	28.2

Dimensioning of valve

When ordering the safety valve you should follow this principle:

Max. admissible tank pressure > set pressure > operating pressure

You should as a minimum have a 10% buffer between the maximum admissible tank pressure and the set pressure.

The set pressure should as a guideline be 20% higher than the expected operating pressure of your process. This is to avoid opening the valve during normal operation.

Please note from the illustrations below here, that if the valve has been opened, the process pressure needs to be reduced significantly to ensure that it has fully closed again.

Valve characteristics when opening and closing

These two following illustrations shows the opening and closing characteristics of the valve for different media.

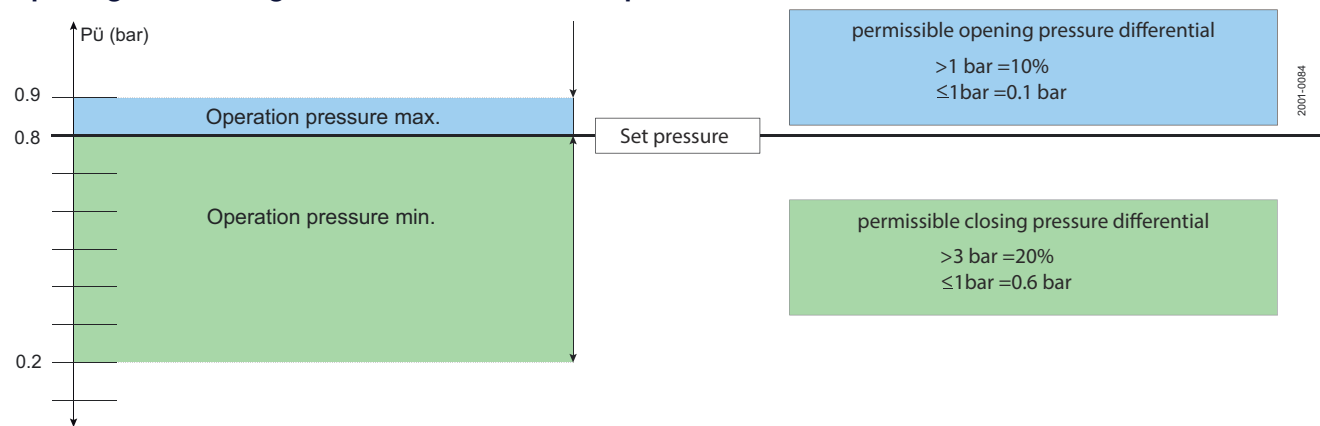


Note!

Note that a 10% buffer above the set pressure is required to ensure that the valve is fully open.

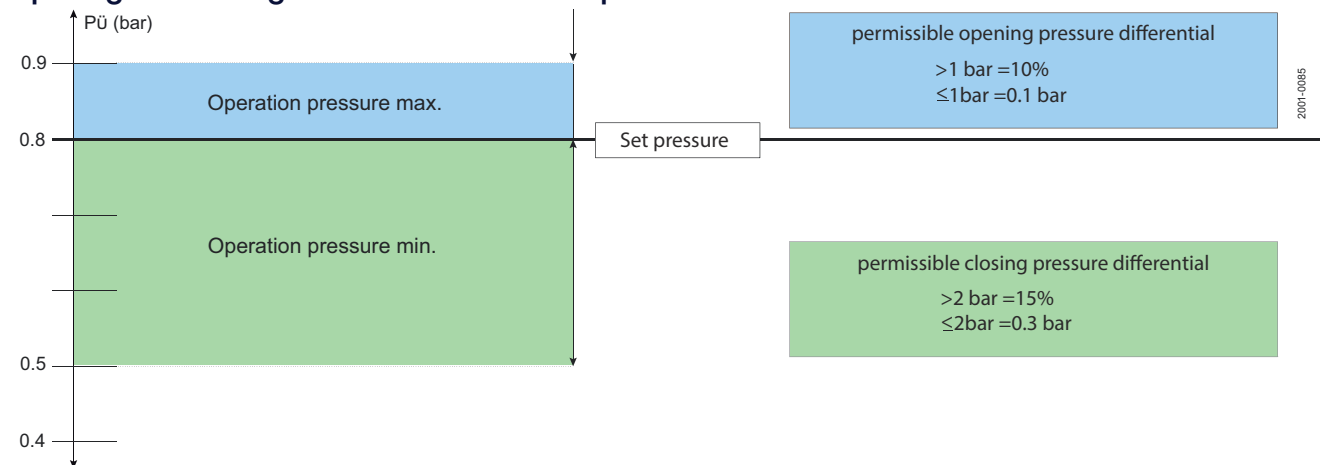
After opening of the valve, the pressure needs to be reduced significantly below the set pressure to ensure that the valve is fully closed again.

Opening and closing characteristics for incompressible fluids



(example: set pressure = 0.8 bar)

Opening and closing characteristics for compressible fluids



(example: set pressure = 0.8 bar)

Blow-off performance chart

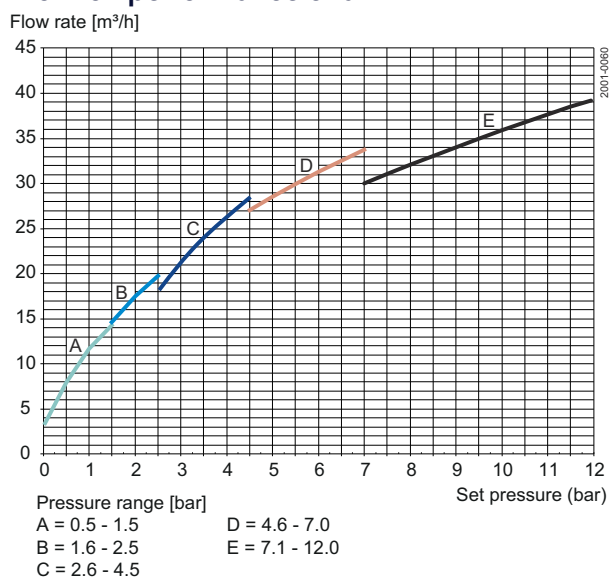


Figure 9. DN25 set pressure: 0.2 - 12.0 bar for liquids (water 20 °C)

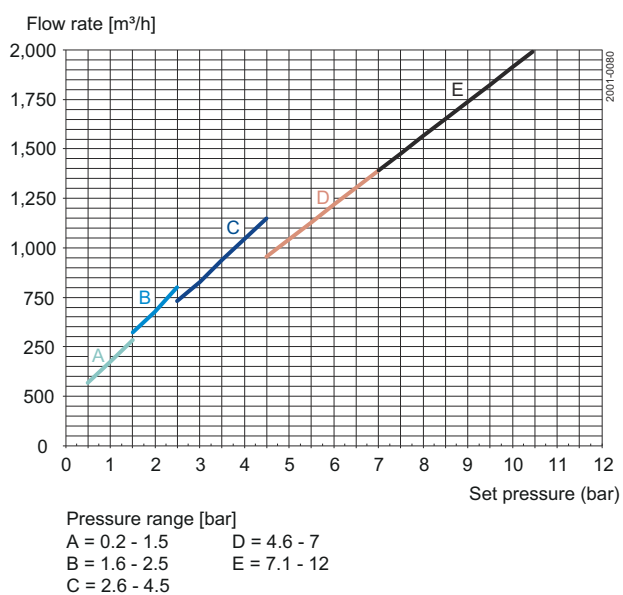


Figure 10. DN 25 set pressure: 0.2 - 12 bar for gases (air 20 °C)



Note!

DN25 for gas application up to 1,5 bar fulfills the DIN4126-1 requirements. For higher pressures the valve is approved by TÜV.

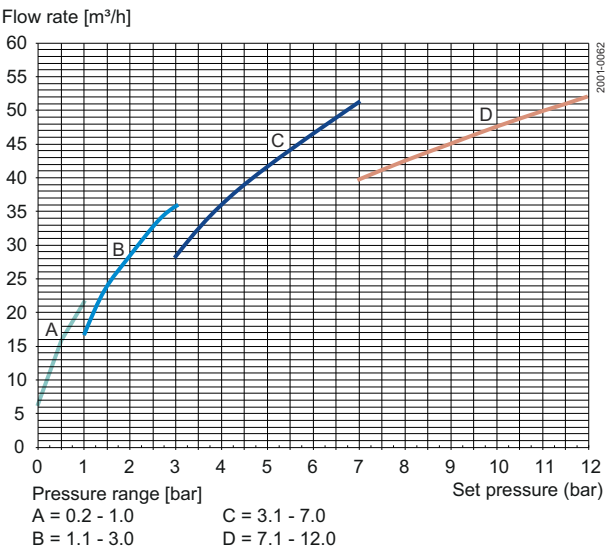


Figure 11. DN 40 set pressure: 0.2 - 12.0 bar for liquids (water 20 °C)

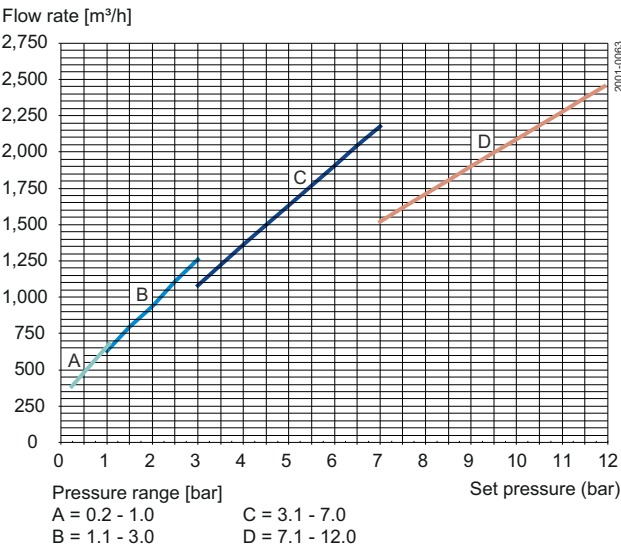


Figure 12. DN 40 set pressure: 0.2 - 12.0 bar for gases (air 20 °C)

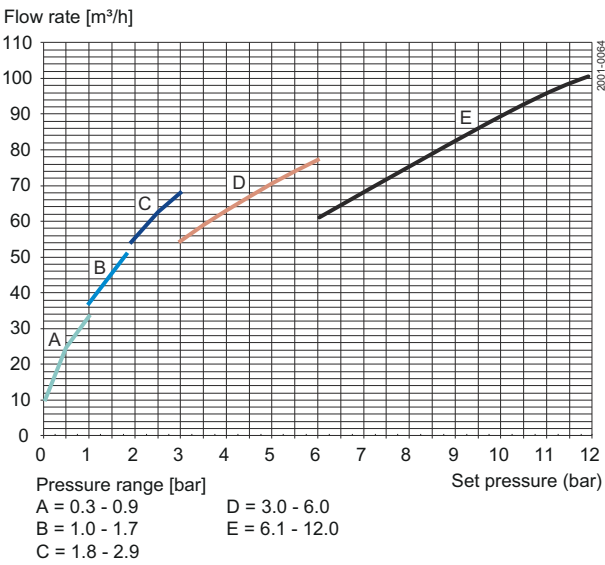


Figure 13. DN 50 set pressure: 0.3 - 12.0 bar for liquids (water 20 °C)

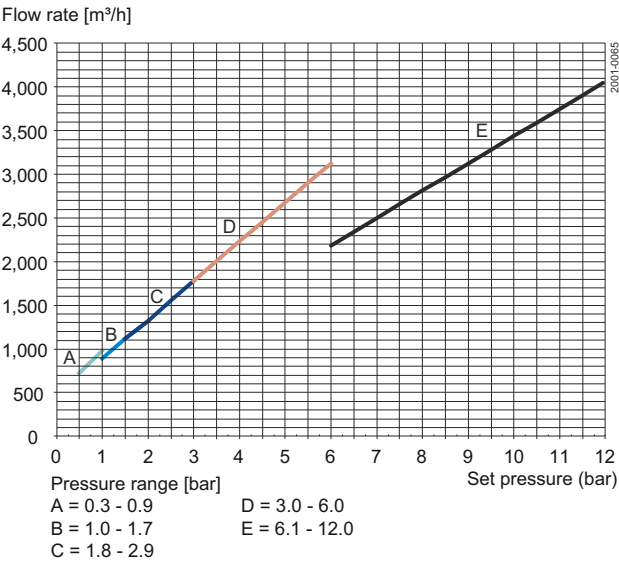


Figure 14. DN50 set pressure: 0.3 - 12.0 bar for gases (air 20 °C)

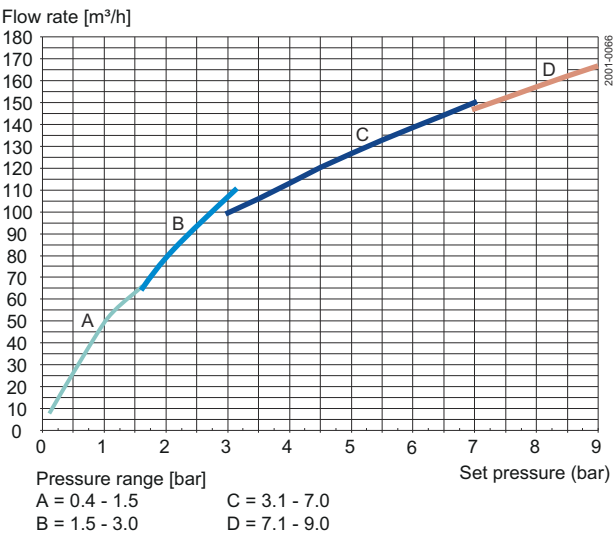


Figure 15. DN65 set pressure: 0.4 - 9.0 bar for liquids (water 20 °C)

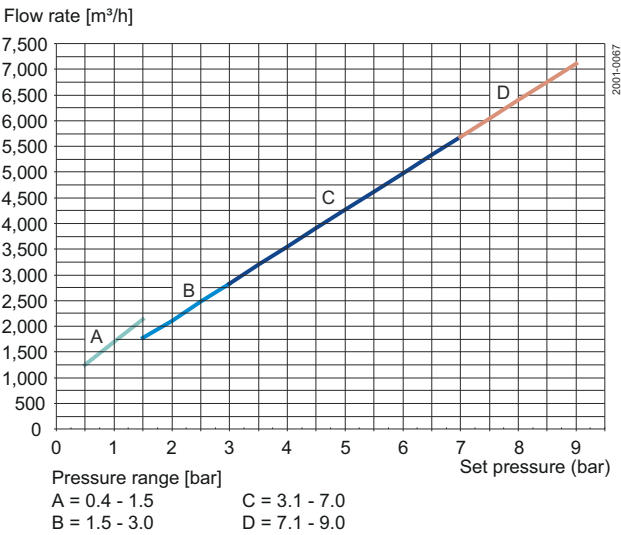


Figure 16. DN65 set pressure: 0.4 - 9.0 bar for gases (air 20 °C)

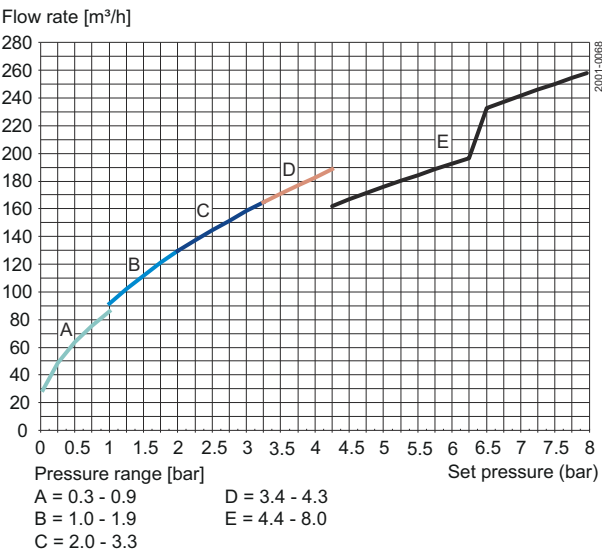


Figure 17. DN80 set pressure: 0.3 - 8.0 bar for liquids (water 20 °C)

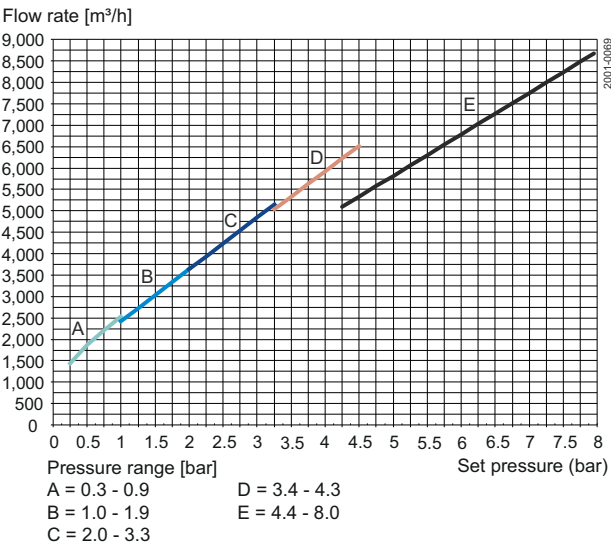


Figure 18. DN80 set pressure: 0.3 - 8.0 bar for gases (air 20 °C)

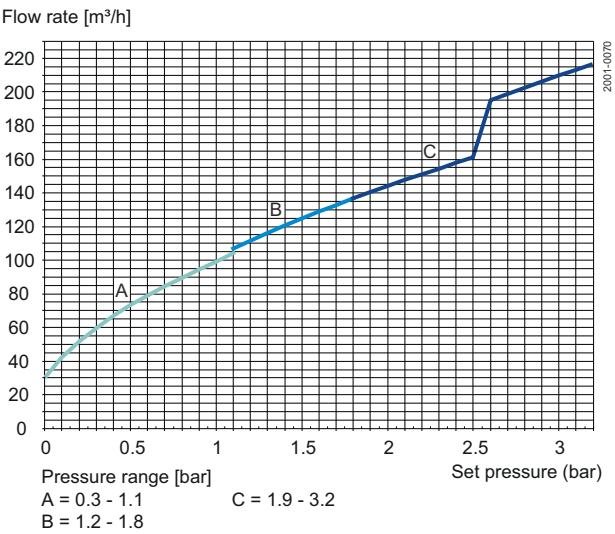


Figure 19. DN100 set pressure: 0.3 - 3.2 bar for liquids (water 20 °C)

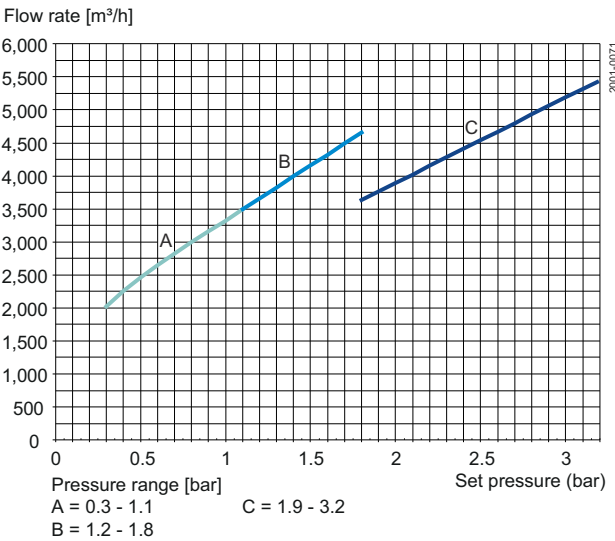


Figure 20. DN100 set pressure: 0.3 - 3.2 bar for gases (air 20 °C)

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